How can public transport stop closures be optimized in Helsinki to minimize passenger disruption using boarding data and simulated rerouting?

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Abstract :

Helsinki, Finland needs to close three of their public transport stops because of constraints in public service funding. This report analyses the stop closure strategy we implemented, which aims to minimize the negative impact on passengers, reduce the amount of people affected and takes the existence of nearby important infrastructure into consideration. It uses open data of the average weekly amount of passengers per stop ID in central Helsinki. As a result, eight under-utilized candidate stops are identified, of which the three stops "Mustikkamaa", "Rakuunantie 8", "Heikinniementie" are selected to be closed with the least impact on vulnerable user groups and an average added delay $\leq$ 1.0 min.

1 Introduction

Public transportation is an important part of the daily lives of Helsinki citizens, with a total of 172 million journeys made in 2024. That being 24% of all journeys in the city, considering transport such as bicycles, cars, and walking trips (City of Helsinki, 2025b; City of Helsinki, Urban Research and Statistics, 2024).
To keep services running smoothly, municipalities must make substantial investments. In 2024, Helsinki spent a total of EUR 87 million on major transport projects, down from EUR 144 million in 2023 (City of Helsinki, 2025a; 2025b). One reason for the expenditures is their investments in new projects such as the tram line connecting the Kalasatama district with the Pasila district (City of Helsinki, 2025b).
In addition to these investments, government budgets are leading to cuts in public service funding, so it is important to rethink existing infrastructure. Three stops must be closed. Consequently, the goal is to identify existing public transport stops whose closure would minimize adverse impacts on passengers so that cost savings are achieved with minimal disruption to regular service. In this context, a stop refers to a single, distinct boarding location with a unique coordinate (and stop_id in the GTFS data set[1]), not the general stop name, which may include multiple such individual stops (stop_name in the GTFS).
It is important to detect what aspects of public transport are essential for passengers. These key elements should guide the closures with the passenger's needs in focus.

1.1 Passenger Needs in Public Transportation

To address the needs of the potentially affected population, it is important to accommodate them as much as possible and cause as little disruption as necessary.

A key concern for public transport users is the time spent waiting and walking, as summarized by Wardman et al. (2014) based on studies in London, 2013. However, as Brazil's data show, perceived safety can moderate the influence on perception of users when walking or waiting between stops (Ruiz-Padillo & de Oña, 2024). The findings suggest that user satisfaction is related to having "adequate conditions for sidewalks, bus stops, public lighting, and the built environment in general" (Ruiz-Padillo & de Oña, 2024, p.14).
In this context, Helsinki benefits from high levels of public safety. Finland was ranked 13th on the Global Peace Index 2024 with a score of 1.474 points, compared to Brazil's 131 with 2.589 points (Institute for Economics and Peace, 2024). The scale ranks countries from 1 the most peaceful and 5 the least peaceful.

[1] GTFS is short for General Transit Feed Specification, a central documentation platform

Furthermore, the count of intentional homicides between 2015 and 2023 in Finland had a mean M = 1.38 (SD = 0.23), which was lower than in Brazil M = 23.87 (SD = 3.76)(United Nations Office on Drugs and Crime, 2025). These indicators suggest that although walking (to the new nearest stop now) and waiting time will slightly increase due to the closures, the safe environment in Helsinki helps minimize its impact. However, additional walking or waiting time should still be minimized.

1.2 Additional Needs

1.2.1 Needs of young population From the research conducted by Broberg and Sarjala (2015), 34% of journeys to and from school have been found to be made by using public transport. With this in mind, when closing stops, we will consider the stop's proximity to schools, as these are essential for children who cannot drive.

1.2.2 Supermarkets/Shopping Salonen et al. (2014) found that in a set of trips where 29.5% were related to shopping and 17.3% to running errands. This influenced our judgment to also consider a stop's proximity to shopping centers and or supermarkets when deciding whether to close a stop.

1.2.3 Needs from Senior Citizens Furthermore, it is important to ensure that the stop closures will not negatively affect elderly people. They make up 18% of the inhabitants in 2023 in Helsinki (City of Helsinki, Urban Research and Statistics, 2024). As a vulnerable group in society, they need to have access to public transport to stay connected and reach essential services, especially since many no longer drive (Karjalainen et al., 2023). Because they often cannot walk longer distances anymore, it is important that closures will not occur in popular areas of elderly citizens. Konečnỳ et al. (2019) denote that the focus on the elderly population is getting more important as the number of senior passengers continues to grow.

1.3 Differences between City Center and Outer City Areas

The degree of dependence on cars is greater in rural areas, where public transport is rare, and most trips are too long to walk or cycle (Pucher & Renne, 2004). Most people who live in rural households own at least one car because they need it as public transport is not fully available for them. Even seniors aged 85 or more covered 51% more mileage per day compared to their urban counterparts. Moreover, the city center area in Helsinki has higher population density in comparison to the outskirts (Albacete et al., 2017), which supports more efficient public transport and makes walking and cycling more feasible.

From this we can conclude that closures should take place in the city center. Although most people living in rural areas are more likely to use private over public transport, we want to maintain a good coverage of all areas within our transport network, while removing stops that will not greatly affect overall accessibility. In addition, these outer-city areas (suburban or rural) have a lower population density, but for some residents losing a stop may mean losing their only connection to essential services. As more services are more likely to appear in urban zones, the closure will be in densely serviced areas, in order to maintain expansive coverage with the transport network. This also means redundant stops can be closed and their budget can be reallocated to more used services within Helsinki.

It is also important to note that Karjalainen et al. (2023) found that pensioners make up the second largest group of vehicle owners within Helsinki. Combined with the fact that there is a high amount of pensioners living in the Helsinki City Center Area (around 20 - 30%) according to Albacete et al. (2017), this only further reinforces our strategy of restricting closures to the city center.

Also, Fink et al. (2024) analyze how fast the central station can be reached using public transport. In order not to worsen the accessibility of the train station and thus connections throughout Finland, it is another argument to limit the possibility of stop closure to the city center. Furthermore, this analysis shows that in addition to the city center - although there are more improvements- there are changes, made through the extension of the tram line, which increase the travel time in the suburban or rural parts of Helsinki (Fink et al., 2024). Therefore, this is another reason why the decision of the closed stops should not be dependent on changes in suburban or rural parts of Helsinki.

1.4 Research Question

Therefore, this follows to the research question which is *How can public transport stop closures be optimized in Helsinki to minimize passenger disruption using boarding data and simulated rerouting?* which will be evaluated by finding the stops through simulation which have the lowest impact and are not needed by the part of the population dependent on public transport. The closure will be in the inner city to not affect the journeys from or to the outer city areas.

2 Methods

2.1 Open Data

For Helsinki openly published data is available on the Helsinki Region Infoshare website (HRI) [2]. Using both the website and the published GTFS data sets as input to our Dijkstra-based routing engine, we created a model to identify the stops that can be closed with the least impact on public transportation users (Helsinki Region Transport Authority (HSL), 2016). We accessed data from the HRI Website which contains the average number of passengers boarding at public transport stops per day in Helsinki from November 2016 (Helsinki Region Transport Authority (HSL), 2016). It is calculated based on travel card validations (corrected to count for travelers who use a pram or a one-time ticket) and includes all types of public transportation.

2.2 Data Preprocessing

To simplify the analysis, the GTFS data set is reduced and specified to services running on Mondays between the 23.05.2025 and the 15.06.2025. Each stop name has as many stop ID's as distinct stops.
The data set in KML format contains the coordinates and Finnish names of transit stops. However, these names do not align with the `stop_name` entries in the GTFS data set, and the coordinates also differ slightly. To reconcile this, a spatial matching process was employed.
Each stop in the GTFS data set was matched with the closest stop in the KML data within a 20-meter radius using geographic proximity. All matched stops were compiled into a single JSON file, which included each stop's ID, name, GTFS coordinates, and average number of passengers. This preprocessing step ensured consistent and spatially accurate data integration across sources.

2.3 Experiment

The entire selection can be traced by following the flow chart 1.

2.3.1 Stop Closure Candidate Selection To limit the analysis to the city center of Helsinki, we manually defined a radius of the capital that includes only coordinates within a diameter of 7 km from the city center. The exact coordinates used for this polygon can be found in the *Appendix A*. Only stops located inside this polygon were considered for further analysis regarding potential closure candidates.

The entire process is summarized in the flowchart in Figure 1.
To evaluate the impact of low-usage public transport stops, 100,000 journeys were simulated using the Halton sequence. The Halton sequence generates evenly distributed points in space, ensuring that the simulation considers a wide and representative range of origin-destination combinations.
Stops were considered as potential candidates for closure if they met all the following criteria: trips made from these stops in the sampled journeys were in the bottom 25% of the frequency table, they were located within the defined city center polygon, and their average passenger count was around 30% of the data.
For each identified stop, the journeys that fulfilled this property were extracted and stored in a CSV file (stop_id, origin and destination coordinates, departure time, the route taken, total travel time).
To assess the impact of closing these candidate stops, the extracted journeys were re-run with the corresponding stop removed from the network. The resulting changes in travel time were calculated for each journey.

In addition, we check that the stops are not next to Assisted (Senior) living facility, hospitals, schools or police stations.

3 Results

To address the research question, boarding data needed to be tested with (re)routing simulations to produce quantifiable results.

[2] Region Infoshare website: https://hri.fi/en_gb/

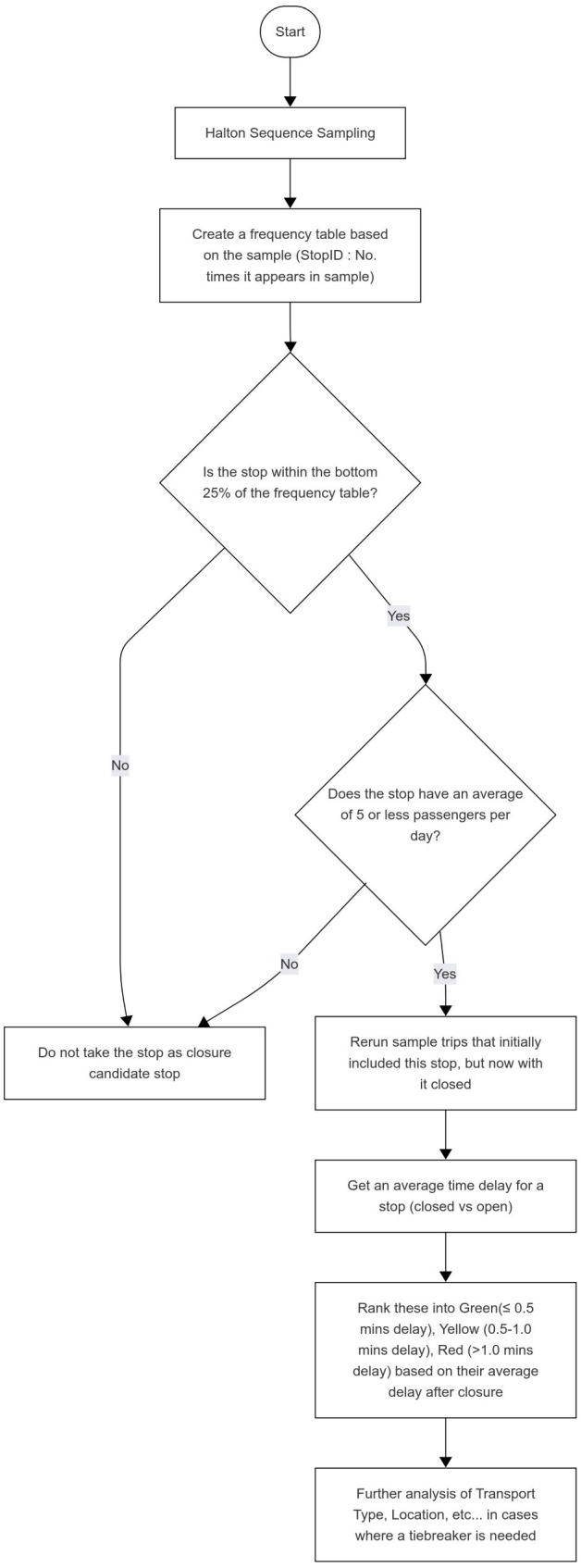


Figure 1. Process of finding optimal stops to close

The cleaned data set did not contain all stops in Helsinki because the KML file was not complete, and therefore it only considered 5637 out of 8313 listed stops in the GTFS (67.8%). As explained above, we further limited the amount of stops to 662, using the inner city polygon approach.

3.1 General results for candidates

The data set generated using the Halton sequence had the number of boardings for each stop for 100,000 trips. Figure 3 shows the resulting stops sorted by their boarding frequencies. The "boarding frequency" is the number of times a given stop was used during the simulation runs. This is not to be confused with the passengers count from the KML file: Here, it indicates the frequency of journeys that use the given stop ID.

The following stops were identified as candidates for closure and can be found in Figure 4, as their boarding frequency falls in the bottom 25% of all stops (see Figure 3 and additionally has an average passenger count of 30% of the exact HRI data. To simplify data, we rounded this threshold to include all stops with five or less average passengers.

Specifically, stop 1301113 was used 52 times in the sample, stop 1301160 was used 39 times, and stop 1320199 was used 26 times. Stops 1420113, 1420121, and 1420125 were each used 13 times and stops 1150128, and 1240112 each 12 times.

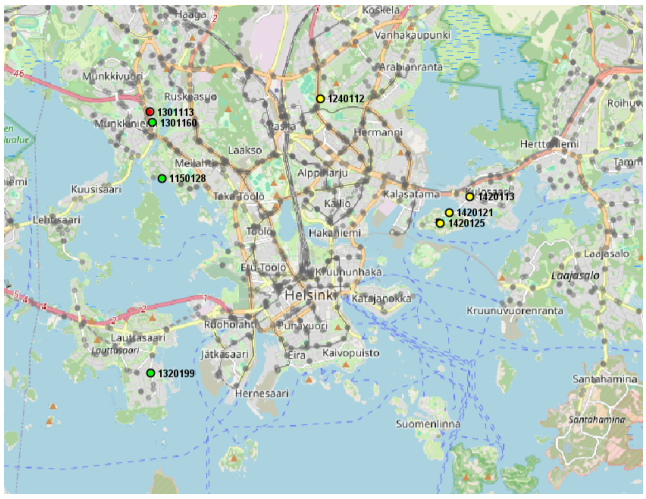


Figure 2. Time impact visualized on the map of Helsinki: per candidate stop color coded by travel delay: Green: ≤ 0.5 min, Yellow: 0.5 – 1.0 min, Red: > 1.0 min

When rerunning the algorithm of the trips, the fol-

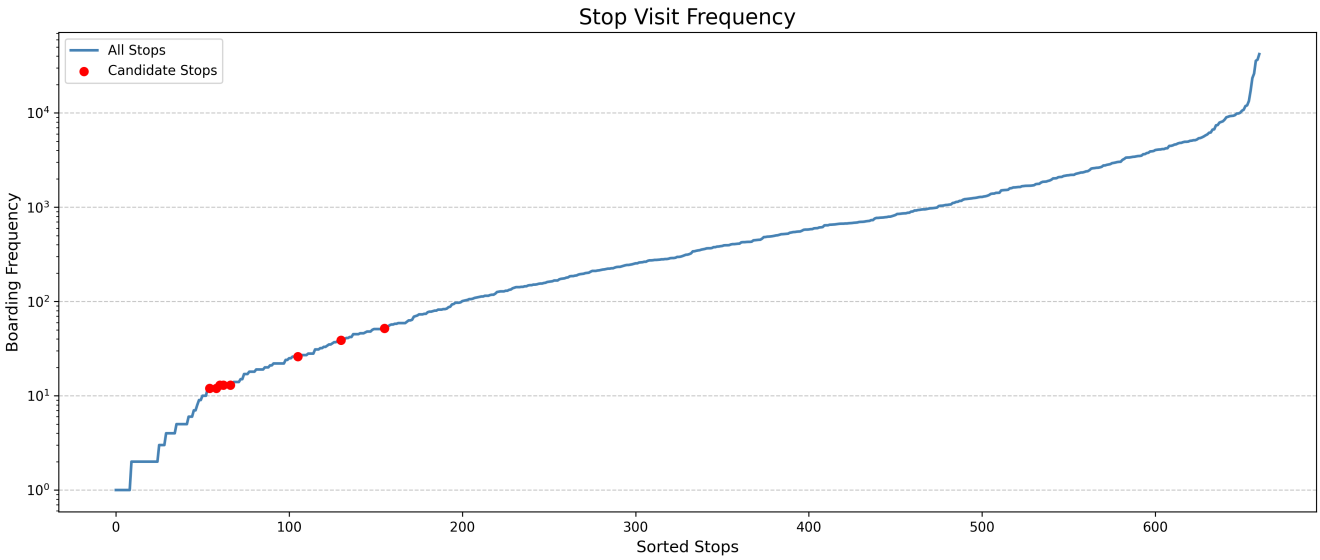


Figure 3. Count of boarding frequency for all stops, in ascending order

lowing time differences were found: Stop 1301113 1.10 Minutes, 1240112 1 Minute 1420113, 1420121, 1420125 0.77 minutes (see Figure 2). They were categorized in the following color categories: Green: ≤ 0.5 min, Yellow: $0.5 - 1.0$ min, Red: > 1.0 min.

3.2 Manual Check of Surroundings

A manual inspection of the surroundings of candidate stops was conducted to identify proximity to significant infrastructure or services (see Table 1).

Stop ID	Surroundings
1301113	No notable infrastructure nearby, very close to 1301160
1301160	No notable infrastructure, very close to 1301113
1320199	Primary school and supermarket < 100m
1420113	Primary school < 500m
1420121	350m walking from 1420125, no significant infrastructure.
1420125	350m walking from 1420121, no significant infrastructure.
1150128	No important infrastructure nearby
1240112	Close to sports facilities, stadiums, and exhibition grounds.

Table 1
Manual check of surrounding infrastructure for candidate stop closures.

4 Discussion

Based on the results of the simulation and analysis of the passenger boarding data, delays, and trip counts, eight candidate stops are identified. Three optimal stops must be selected from these in order to answer the research question.

To ensure that these three stops are the stops that can be closed with limited disruption for passengers of Helsinki's public transport, we applied rigid criteria. Our selected stops must fall in the bottom 25% of the frequency of trips in the 100,000 samples of the Halton sequence simulation, because this captures the most underutilized stops while ensuring a manageable pool for further analysis. The stops also have an average passenger amount of fewer than six per day. The stops with ≤ 6 average daily passengers represent the lowest 30% of ridership in the data set. This cutoff was chosen because these stops serve fewer than one passenger per hour (assuming 12-hour service), making them economically inefficient to maintain relative to their utility.
There are eight lowest-performing stops, which we considered as candidates to be closed.

Three of them received a 'green' score, indicating that their removal has an average delay of less than 0.5 minutes. This delay exists because the shortest trip cannot use the closed stops anymore. This helps quantify the potential cost, in terms of travel delay, of closing underutilized stops and ensures that all users



Figure 4. Candidate stops sorted by boarding frequency and colored according in their delay: Green: ≤ 0.5 min, Yellow: $0.5 - 1.0$ min, Red: > 1.0 min

of public transportation in Helsinki are minimally affected.

The following are the three green stops that we considered first for removing: "Rakuunantie 8" (GTFS stop_id: 1301160), "Vattuniemenkuja" (GTFS stop_id: 1320199) and "Heikinniementie" (GTFS stop_id: 1150128).

Manual checks confirmed that Stop 1301160 and 1150128 are not located near essential infrastructure such as schools, shops, living facilities. Thus, their closure has the minimal delay and is unlikely to negatively affect vulnerable users of Helsinki's public transport (Broberg & Sarjala, 2015; City of Helsinki, Urban Research and Statistics, 2024; Karjalainen et al., 2023; Salonen et al., 2014).

However, "Vattuniemenkuja" (Stop 1320199) is in close proximity to a primary school and a supermarket (< 100m). Therefore, the third removal needs to be further investigated.

We proceeded to look at the four 'yellow' score stops that had an average delay of $0.5 - 1.0$ min for all the trips considered. These four stops had a boarding frequency between 12 and 13 as a result of the simulation. The stop with the lowest frequency is "Sofianlehto" (1240112). Despite the low score, we identified multiple sports facilities, olympic stadiums, and an exhibition ground in close proximity of the stop. As our analysis took place on a Monday, it is reasonable that the data set does not include the extraordinary events that usually take place in these public spaces. However, when such events take place, "it is very important that business tourists reach the event venue by public transports" (Universities, 2015, p. 52). Therefore, we decided not to close this stop. Stop "Kulosaarentie" (1420113) has a primary school in a 500 m proximity and was therefore not favorable for closure as both children and teachers need to be accommodated by public transport Broberg and Sarjala (2015). We also considered the stops "Mustikkamaa" (1420121) and "Mustikkamaanpolku" (1420125) since they are in a 350 m walking distance to each other and no notable infrastructure was identified nearby. "Mustikkamaa" is placed between two adjacent stops, while "Mustikkamaanpolku" is the final destination for the bus trip. Therefore, we chose to close "Mustikkamaa" (1420121) as the user would still benefit from two other walkable options.

Hence, the selected stops to be closed are "Mustikkamaa", "Rakuunantie 8", "Heikinniementie" and the 'red' labeled stop did not need to be considered.

As previously stated, this analysis only considers stop closures in the city center of Helsinki. This re-

striction ensures that all areas of Helsinki retain access to public transport through stops. By limiting the analysis to centrally located stops with low passenger frequency ($\leq$ 5 passengers daily), we avoid impacting long-distance journeys which rely on rare but essential trips.

Furthermore, to support vulnerable groups, such as children, seniors and sick people in Helsinki, we manually checked that no hospitals, schools, or living facilities were found in the immediate surroundings of closed stops. While the closure of selected stops may slightly increase walking and waiting times, Helsinki remains a safe environment that makes these changes manageable for most healthy passengers (Ruiz-Padillo & de Oña, 2024; Wardman et al., 2014).

All our proposed closures are bus stops. While the exact economical savings from removing these stops cannot be quantified within this report, it is clear that eliminating underused infrastructure saves energy and maintenance costs without affecting the overall service quality. In 2023, bus trips made up 25,4% of all public transport journeys in Helsinki (63 million of 248 million), stressing the importance of optimizing the bus sector (City of Helsinki, Urban Research and Statistics, 2024).

Finally, while this report identifies inefficient parts of the public transport service system, it also supports broader environmental goals. Helsinki continues to invest in new public transport projects, such as creating new tram lines, which is essential for sustainable urban development (City of Helsinki, 2025a; 2025b; Santos et al., 2021). These stop closures aim to evaluate and improve inefficient aspects of the existing infrastructure.

4.1 Limitations and future research

A key limitation of our research on closures is the restrictions we applied to get the list of stops considered. Firstly, only data in a self-determined polygon was used, meaning that it cannot be fully determined that the three optimal stops have been found. Furthermore, due to inconsistencies between the stops of the GTFS and KML datasets, 67.7% of Helsinki's stops were matched, and therefore partly used in the polygon.

Furthermore, our analysis did not consider the spacing of the bus stops across Helsinki, except for the specific case of the closure of stop 1420121, where we inferred that people could instead use stop 1420125. Future research should investigate further the spatial spacings of stops, as this is an option to find removable stops (Johar et al., 2017).
Another limitation is the use of outdated passenger average data, as it is from November 2016. We still managed to extract meaningful insights from the dataset, but future research should conduct comparative analysis for accuracy.

5 Conclusion

Within this report, we analyzed the 662 stops in Helsinki's city center by combining boarding open data counts and simulating with the Halton sequence. This approach led us to find the optimal stops to close so that public transport users are minimally affected. Three stops ("Mustikkamaa", "Rakuunantie 8", "Heikinniementie") are chosen to be closed with an average added delay $\leq$ 1.0 min and minimal effect on vulnerable user groups.

Our findings demonstrate that stop closures can lead to operational savings and minimal rider disruption, while still addressing the needs of vulnerable passengers.

Future research should integrate more recent boarding data and develop a focus on stops' proximity spacing.

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6 Appendix

6.1 Appendix A

The exact coordinates used for the Polygon are the following seven:

```
(60.1952476629647, 24.88824790574304),
(60.20224721136488, 24.963022712395038),
(60.18665854871923, 24.982188294283407),
(60.16672892559488, 24.98001038729288),
(60.14722085100674, 24.921787673490083),
(60.15112339211973, 24.903783642368403),
(60.166078842714306, 24.89826627799241)
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